1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

ANS:- Main Components Inside a Desktop PC

Component	Description
Motherboard	The main circuit board that connects and allows communication between all computer components.
CPU (Processor)	The brain of the computer that performs calculations and executes instructions.
RAM (Random Access Memory)	Temporary memory used to store data and programs that are currently in use.
Storage Drive (HDD/SSD)	Stores the operating system, software, and user files permanently. SSDs are faster than HDDs.
SMPS (Power Supply Unit)	Converts AC electricity into DC power and supplies power to all components.
Graphics Card (GPU) (if installed)	Processes graphics and video output, especially useful for gaming and graphics-intensive tasks.
SATA Data Cables	Connect HDDs, SSDs, and optical drives to the motherboard.
Power Cables	Carry electrical power from the SMPS to the motherboard, drives, and other hardware.
Cabinet (Case)	Houses and protects all internal computer components.
Cooling Fans	Improve airflow inside the cabinet and help keep components cool.
CMOS Battery	Maintains BIOS settings and the system clock when the PC is powered off.

2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step. Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

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3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Step	What I Connected/Installed	Why the Order Matters
Step 1: Install the Motherboard	Placed the motherboard inside the cabinet, aligned it with the I/O shield and standoffs, and secured it with screws.	The motherboard is the main platform that holds and connects most other components, so it must be installed first.
Step 2: Install the CPU	Carefully aligned the CPU with the socket, placed it into position, and locked the socket lever.	The CPU should be installed before the cooler because the socket is easily accessible.
Step 3: Install the CPU Cooler	Applied thermal paste (if required), mounted the CPU cooler, tightened the screws, and connected the CPU fan cable to the motherboard.	The cooler must be installed immediately after the CPU to prevent overheating and because it is easier before other components are installed.
Step 4: Install the RAM	Inserted the RAM modules into the correct DIMM slots until the retaining clips clicked into place.	RAM installation is easier before larger components, such as the graphics card, block access.
Step 5: Install the HDD/SSD	Mounted the HDD/SSD in its drive bay and connected the SATA data cable to the motherboard and the SATA power cable from the SMPS.	Storage devices must be connected before cable management to ensure all cables are properly routed.
Step 6: Install the Graphics Card (GPU)	Inserted the graphics card into the PCIe x16 slot, secured it with screws, and connected the PCIe power cable (if required).	The GPU is installed after the motherboard, CPU, RAM, and storage because it is large and can obstruct access to other components.
Step 7: Connect Power Supply Cables	Connected the 24-pin ATX motherboard power cable, 8-pin CPU power cable, SATA power cables, and GPU power cable.	All components require power before the system can start, so these connections are completed after the hardware is installed.
Step 8: Connect Front Panel Cables	Connected the power switch, reset switch, power LED, HDD LED, USB, and front audio cables to the motherboard headers.	These connections enable the cabinet buttons, LEDs, USB ports, and audio ports to function correctly.
Step 9: Organize Cables and Close the Cabinet	Arranged cables using cable ties, ensured nothing blocked the fans, and reattached the side panel.	Proper cable management improves airflow, cooling, and makes future maintenance easier.

Step 10: Test the System	Connected the monitor, keyboard, mouse, and power cable, then switched on the PC and verified that it booted successfully.	Testing confirms that all components have been installed and connected correctly before regular use.

4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

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Tools Needed

Tool	**Why It Is Important**
Screwdriver	Used to remove and tighten most PC screws.
Anti-Static Wrist Strap	Prevents electrostatic discharge (ESD) from damaging components.
Soft Brush	Removes dust safely from components.
Compressed Air	Cleans dust from hard-to-reach areas.
Cable Ties	Keeps cables organized and improves airflow.
Small Container	Stores screws safely to prevent loss.

Safety Precautions

Precaution	**Importance**
Turn off the PC completely	Prevents electrical accidents.
Unplug the power cable	Eliminates the risk of electric shock.
Handle components by the edges	Prevents damage to chips and circuits.
Avoid touching CPU pins or motherboard contacts	Reduces the risk of bending pins or causing electrical damage.
Work on a clean, dry surface	Prevents dust, moisture, and accidental short circuits.
Keep screws organized	Prevents losing screws and ensures proper reassembly.
Do not force components into slots	Prevents physical damage to connectors and sockets.